# Mth631

## Q-1

Suppose $U_0 \in T$ is a limit point of $T$, $G$ is continuous at $U_0$ and $f$ is continuous at $X_0 = G(U_0)$. Then, show that $f \circ g$ is continuous at $U_0$? Mark: 10

**Proof:-**

Suppose $\varepsilon > 0$. Since $f$ is continuous at $X_0 = G(U_0)$, there is an $\varepsilon_1 > 0$ such that

$$|f(X) - f(G(U_0))| < \varepsilon \longrightarrow (1)$$

if

$$|X - G(U_0)| < \varepsilon_1 \text{ and } X \in D_f \longrightarrow (2)$$

Since $G$ is continuous at $U_0$, there is a $\delta > 0$ such that

$$|G(U) - G(U_0)| < \varepsilon_1$$ M.M

if $|U - U_0| < \delta$ and $U \in D_G$.

by taking $X = G(U)$ in (1) and (2) we get,

$$|h(U) - h(U_0)| = |f(G(U)) - f(G(U_0))| < \varepsilon$$

if

$$|U - U_0| < \delta \text{ and } U \in T$$

**Note:-**

"$U$" is a sign in Mathtype.

$U_0$

**Q.2**

Show that $(0,0)$ is a critical point of the function $p(x,y) = x^2 - 2xy + y^2 - x^4 - y^4$ and it has positive semidefinite second differentials at $(0,0)$?

**Soln:-**

Given $p(x,y) = x^2 - 2xy + y^2 - x^4 - y^4$

$p_x(x,y) = 2x - 2y - 4x^3$

$p_y(x,y) = -2x + 2y - 4y^3$

M.M

* At point $(0,0)$

$p_x(0,0) = 0$ and $p_y(0,0) = 0$

So, $(0,0)$ is a critical point of $p$.

Now,

$p_{xx}(x,y) = 2 - 12x^2$

$p_{yy}(x,y) = 2 - 12y^2$

$p_{xy}(x,y) = -2$

$(d^2_{(0,0)} p)(x,y) = 2x^2 - 4xy + 2y^2$

$(d^2_{(0,0)} p)(x,y) = 2(x^2 - 2xy + y^2)$

$= 2(x-y)^2$

Hence if is +ve semidefinide.

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**Q.3**

Show that $(0,0)$ is a critical point of the function $p(x,y) = x^2 - 2xy + y^2 + x^4 + y^4$ and it has +ve semidefinite second differentials at $(0,0)$.

**Soln:-**

Given

$$p(x,y) = x^2 - 2xy + y^2 + x^4 + y^4$$

$$p_x(x,y) = 2x - 2y + 4x^3$$

$$p_y(x,y) = -2x + 2y + 4y^3$$

At point $(0,0)$

$p_x(0,0) = 0$ and $p_y(0,0) = 0$

So $(0,0)$ is a critical point of $p$.

Now

$$p_{xx}(x,y) = 2 + 12x^2$$

$$p_{yy}(x,y) = 2 + 12y^2$$

$$p_{xy}(x,y) = -2$$

M.M

$$(d^2_{(0,0)}p)(x,y) = 2x^2 - 4xy + 2y^2$$

$$(d^2_{(0,0)}p)(x,y) = 2(x^2 - 2xy + y^2)$$

$$(d^2_{(0,0)}p)(x,y) = 2(x-y)^2$$

Hence it is +ve semidefinite.

## Q.4

Define a locally integrable function. Also show that the function $f(x) = \log x$ is locally integrable on $[0,1)$.

**Soln:-**

Locally integrable function.

If $f$ is a function, it is locally integrable on an interval $I$ if $f$ is integrable on every finite closed subinterval of $I$.

Proof:-

$$\int_1^0 \log x\,dx = x\ln x - x\Big|_1^0$$ M.M

$$= 0 - 1$$

$$= -1$$

Because it is unbounded as $x \to 0+$

since

$$\lim_{c\to 0+}\int_c^1 \log x\,dx = \lim_{c\to 0+}(x\log x - x)\Big|_c^1$$

$$= -1 - \lim_{c\to 0+}(c\log c - c) = -1$$

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**Q.5**

Suppose that $f(x,y) = \begin{cases} 0 & \text{if } x \text{ and } y \text{ are rational} \\ 1 & \text{if } x \text{ is rational and } y \text{ is irrational} \\ 2 & \text{if } x \text{ is irrational and } y \text{ is rational} \\ 3 & \text{if } x \text{ and } y \text{ are irrational} \end{cases}$

**Soln:-**

Let $P = \{R_1, R_2, \cdots R_k\}$

$M_j = \sup\{f(x,y) / X \in R_j\}$

$m_j = \inf\{f(x,y) / X \in R_j\}$ M.m

Let $j$ be arbitary in $\{1,2,\cdots k\}$. since $R_j$ contains a point $(\hat{x}_j, \hat{y}_j)$ with $\hat{x}_j$ and $\hat{y}_j$ irrational $M_j = 3$. Hence, $\underline{\int_R} f(x,y)\,d(x,y) = 3(b-a)(d-c)$.

Since $R_j$ contains a point $(\tilde{x}_j, \tilde{y}_j)$ with $\tilde{x}_j$ and $\tilde{y}_j$ rational. $m_j = 0$ Hence,

$$\underline{\int_R} f(x,y)\,d(x,y) = 0$$

**Q.6**

If a nonempty subset $S$ of $R^n$ is both open and closed, then $S = R^n$.

**Soln:-**

Let $X_0$ is in $S$. If $S \neq R^n$, there is an $X_1 \notin S$.

Let $H = \{\tau | (1-t)X_0 + tX_1 \in S \text{ for } 0 \leq 1 < \tau\}$

Since $S$ is open it contains a neighborhood of $X_0$. So, $H \neq \phi$.

since $X_1$ is not in $S$, $\tau \leq 1$ for all $\tau$ in $H$.

let $\rho = \sup H$ and $\overline{X} = (1-\rho)X_0 + \rho X_1$

then $\overline{X}$ is a limit point of $S$ and so in $S$. because $S$ is closed. since $S$ is open. it contains some $\epsilon$—neighborhood of $\overline{X}$, so $\rho + \frac{\epsilon}{2}$ is in $H$. If gives the definition of $\rho$.

Hence, $S = R^n$.

M.M

**Q.7**

Define Linear Function:-

**Ans:-**

A linear function is a form of function

$$L(X) = m_1x_1 + m_2x_2 + \cdots + m_nx_n$$

where $m_1, m_2, \cdots m_n$ are constant.

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**Q.8**

If R is degenerate, then prove by using the definition of Riemann integral that $\int_R f(X)dA = 0$ if f is bounded on R.

**Proof :-**

let

$$R = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n] \text{ with } a_1 = b_1$$

then every partition P of R is of the form

$$P = [R_1, R_2, R_3, \ldots, R_K]. \text{ where } 1 \leq j \leq K.$$

M.M

Therefore,

Every Riemann sum of f over R equals Zero, So

$$\int_R f(X)dA = 0$$

Q.9

Determine whether the integral

$$\int_0^{\infty} \frac{1+\cos^2 x}{\sqrt{1+x^2}}\,dx$$

converges or diverges.

Sol:-

since

$$\frac{1+\cos^2 x}{\sqrt{1+x^2}} > \frac{1}{\sqrt{1+x^2}} = \frac{1}{x\sqrt{\frac{1}{x^2}+1}} > \frac{1}{x\sqrt{2}}$$

if $x > 1$.

since

$$\int_1^{\infty} \frac{dx}{x} = \infty$$

M.m

So,

$$\int_0^{\infty} \frac{1+\cos^2 x}{\sqrt{1+x^2}} = \infty$$

this is diverges.

Q.10

Calculate $\frac{\partial f(X)}{\partial \phi}$ where $f(x,y,z) = \log(1+x$

and $\phi = (0,1,0)$.

Sol:-

We Know that

$$h(t) = f(X + t\phi)$$ M.M

Given

$$f(x,y,z) = \log(1+x+y+z)$$

$$h(t) = f(x+\phi_1 t \cdot y + \phi_2 t \cdot z + \phi_3 t)$$

$$= \log(1+x+y+z+(\phi_1+\phi_2+\phi_3)t)$$

$$h'(t) = \frac{\phi_1+\phi_2+\phi_3}{1+x+y+z+(\phi_1+\phi_2+\phi_3)t}$$

at $t=0$ we get,

$$\frac{\partial f(X)}{\partial \phi} = h'(0) = \frac{\phi_1+\phi_2+\phi_3}{1+x+y+z}$$

Now put $\phi_1 = 0, \phi_2 = 1, \phi_3 = 0$

$$\frac{\partial f(X)}{\partial \phi} = \frac{0+1+0}{1+x++y+z} = \frac{1}{1+x+y+z}$$

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Q.11

Prove that the function $f(x,y) = 3x^2y^3 + xy$ has eight third-order partial derivatives.

Sol:

Given $f(x,y) = 3x^2y^3 + xy$

1st-order partial derivatives

$f_x(x,y) = 6xy^3 + y$

$f_y(x,y) = 9x^2y^2 + x$

2nd-order partial derivatives

$f_{xx}(x,y) = 6y^3$

$f_{yy}(x,y) = 18x^2y$

$f_{xy}(x,y) = 18xy^2 + 1$

$f_{yx}(x,y) = 18xy^2 + 1$

M.M

There are eight third-order derivatives.

$f_{xxx}(x,y) = 0$

$f_{yyy}(x,y) = 18x^2$

$f_{xxy}(x,y) = 18y^2$

$f_{xyx}(x,y) = 18y^2$

$f_{yxx}(x,y) = 18y^2$

$f_{yyx}(x,y) = 18y^2$

$f_{yxy}(x,y) = 18y^2$

$f_{xyy}(x,y) = 18y^2$

## Q.12

Compute $f_{xy}(0,0)$ and $f_{yy}(0,0)$ if

$$f(x,y)=\begin{cases}\dfrac{(x^3y+xy^3)\sin(x-y)}{x^2+y^2} & (x,y)\neq(0,0)\\ 0, & (x,y)=(0,0).\end{cases}$$

**Sol:-**

$$f_{xy}(0,0)=\lim_{y\to 0}\frac{f_x(0,y)-f_x(0,0)}{y}$$

$$=\lim_{y\to 0}\frac{-\sin y-0}{y}=-1$$

$$f_{yy}(0,0)=\lim_{y\to 0}\frac{f_y(0,y)-f_y(0,0)}{y}$$

$$=\lim_{y\to 0}\frac{0-0}{y}=0$$ M.M

## Q.13

Maxima and Minima

**Ans:-**

If $X_0$ is a local extreme point of $f$ if there is a $\delta>0$ such that

$$f(X)-f(X_0)$$

does not change sign in $S_\delta(X_0)\cap D_f$ then Maxima is $f(X)\leq f(X_0)$ and minima is $f(X)\geq f(X_0)$ for all $X$ in $S_\delta(X_0)\cap D_f$

Q.14

Show that the function $f(x,y) = x^2 + 2xy$ is differentiable at point $(x_0, y_0)$.

Sol:-

$$f(x,y) - f(x_0,y_0) = x^2 + 2xy - x_0^2 - 2x_0y_0$$

$$= x^2 - x_0^2 + 2(xy - x_0y_0)$$

M.M

$$= (x-x_0)(x+x_0) + 2(xy - x_0y_0)$$

$$= (x-x_0)(x+x_0) + 2(xy - x_0y + x_0y - x_0y_0)$$

$$= (x-x_0)(x+x_0) + 2(xy - x_0y) + 2(x_0y - x_0y_0)$$

$$= (x-x_0)(x+x_0) + 2y(x-x_0) + 2x_0(y-y_0)$$

$$= (x-x_0)(x+x_0+2y) + 2x_0(y-y_0)$$

$$= 2(x_0+y_0)(x-x_0) + 2x_0(y-y_0) + (x-x_0)(x-x_0+2y-2y_0)$$

$$= m_1(x-x_0) + m_2(y-y_0) + (x-x_0)(x-x_0+2y-2y_0)$$

here $m_1 = 2(x_0+y_0) = f_x(x_0,y_0)$ and $m_2 = 2x_0 = f_y(x_0,y_0)$

[redacted], by Schwarz's inequality

$$\lim_{X \to X_0} \frac{f(x,y) - f(x_0,y_0) - m_1(x-x_0) - m_2(y-y_0)}{|X - X_0|} = 0$$

So, function is differentiable at $(x_0,y_0)$.

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**Q.15**

Define Riemann Sum and Reimann integral in $R^n$.

**Ans:-**

**Riemann sum in $R^n$**

If $f$ is a real-valued function define on a rectangle $R$ in $R^n$ and $P=\{R_1, R_2, \ldots, R_K\}$ is a partition of $R$. let $X_j$ is an arbitrary point in $R_j$ such that $1 \leq j \leq K$ then

M.m

$$\sigma = \sum_{j=1}^{K} f(X_j)\, V(R_j)$$

is a Reimann sum of $f$ over $P$.

**Riemann integral in $R^n$**

If $\sigma$ is any Riemann sum of $f$ over a partition $P$ of $R$ such that $\|P\| < \delta$. here, we can say that $L$ is the Riemann integral of $f$ over $R$. and written as

$$\int_R f(x)\,dx = L$$

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Q-16

The function $f(x,y) = 1 - x^2 - 2y^2$ is continuous on $R^2$.

Proof:-

Give

$f(x,y) = 1 - x^2 - 2y^2$ M.M

if $f(x,y)$ is continuous on $\mathbb{R}^2$ then it is continuous on all points.

So,

$f(x_0, y_0) = 1 - x_0^2 - 2y_0^2$, $x_0, y_0 \in \mathbb{R}^2$

i.e

$f(1,-1) = 1 - (1)^2 - 2(-1)^2$

$= 1 - 1 - 2$

$= 0$

so it is continuous

**Q.17**

Show that the space $R^n$ is connected.

**Proof:-**

If $R^n = A \cup B$ with $\bar{A} \cap B = \phi$ and $A \cap \bar{B} = \phi$, then $\bar{A} \subset A$ and $\bar{B} \subset B$. That is, A and B are both closed and therefore are both open. Since the only nonempty subset of $R^n$ that is both open and closed is $R^n$ itself, one of A and B is $R^n$ and the other is empty.

M.m

**Q.18**

Show that $\lim_{x \to x_0} f(x)$ exists, then it is unique.

**Proof:-**

Let f(x) be a function.

$$\lim_{x \to x_0} f(x)$$

M.m

Then it is unique if the Domain of f(x) is,

$$D_f \subseteq R^n$$

## Q.19

The function $f(x,y) = \frac{xy}{x^2+y^2}$ is defined everywhere in $\mathbb{R}^2$ except at $(0,0)$

Does $\lim_{(x,y)\to(0,0)} f(x,y)$ exist?

**Sol:-**

Let $(x,y)$ approach $(0,0)$ along the line $y=x$, then function values will be

$$f(x,x) = \frac{xy}{x^2+y^2} = \frac{x\cdot x}{x^2+x^2} = \frac{x^2}{2x^2} = \frac{1}{2}$$

If $(x,y)$ approach $(0,0)$ along the line $y=-x$, then function value will be

$$f(x,-x) = \frac{xy}{x^2+y^2} = \frac{x(-x)}{x^2+(-x)^2} = \frac{-x^2}{2x^2} = -\frac{1}{2}$$

it is clear that $f(x,x)$ and $f(x,-x)$ is not equal.

M.M

Now

$\lim_{x\to 0} f(x,x) = \frac{1}{2}$ and $\lim_{x\to 0} f(x,-x) = -\frac{1}{2}$

$\Rightarrow \frac{1}{2} \neq -\frac{1}{2}$

Hence, $\lim_{(x,y)\to(0,0)} f(x,y)$ not exist.

it will exist if $\lim_{x\to 0} f(x,x) = \lim_{x\to 0} f(x,-x)$.

**Q.20**

Show that if $f$ is differentiable at $X_0$, then $f$ is continuous at $X_0$.

**Proof:-**

We know by linear function

$$L(X) = m_1x_1 + m_2x_2 + \cdots + m_nx_n$$

by Schwarz's inequality

$$|L(X-X_0)| \leq M|X-X_0|$$ M.m

where $M = (m_1^2 + m_2^2 + \cdots + m_n^2)^{1/2}$

and

$$f(X) - f(X_0) = L(X-X_0) + E(X)(|X-X_0|)$$

imply that,

$$|f(X)-f(X_0)| \leq (M + |E(X)|)|X-X_0|.$$

when

$$\lim_{X \to X_0} E(X) = 0$$

then $f$ is continuous at $X_0$.

Hence proved.

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Q.21

If $S_1$ has zero content and $S_2 \subset S_1$ then prove $S_2$ has zero content.

If $\epsilon > 0$, there are rectangles $T_1, T_2, T_3, \dots T_r$ and $T_1', T_2', T_3', \dots T_s'$ such that if $S_1 \subset \bigcup_{i=1}^{r} T_i$ and $S_2 \subset S_1$ then

$S_2 \subset \bigcup_{i=1}^{r} T_i$. M.M

Hence $S_2$ has zero content.

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**Q.22**

If $f(s)=\sqrt{s}$ and $g(x,y)=1-x^2-2y^2$ then $D_f=[0,\infty]$, $D_g=R^2$ and $T=\{(x,y): x^2+2y^2\leq 1\}$.

**Sol:-**

since $g$ is continuous on $R^2$

Let

$f(g(x,y))=\sqrt{1-x^2-2y^2}$ and $f(s)=\begin{cases}\frac{\sin s}{s}, & s\neq 0\\ 1, & s=0\end{cases}$

then $D_f=(-\infty,\infty)$ and M.M

$D_g=T=\{(x,y): x^2+2y^2\leq 1\}$.

So the $g$ is continuous on $T$.

since $f$ is continuous on $D_f$, the composite function $h=fog$ defined by

$$h(x,y)=\begin{cases}\frac{\sin\sqrt{1-x^2-2y^2}}{\sqrt{1-x^2-2y^2}}, & x^2+2y^2\leq 1,\\ 1, & x^2+2y^2=1,\end{cases}$$

is continuous on

$D_g=T=\{(x,y): x^2+2y^2\leq 1\}$

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Q.23

rth Differential

Sol:-

If $r \geq$ and all the partial derivatives of $f$ of order $\leq r-1$ are differentiable in a neighborhood of $X_0$.

Then the $r$th differential of $f$ at $X_0$, denoted by $d^{(r)}_{X_0} f$. defined by

M.M

$$d^{(r)}_{X_0} f = \sum_{\gamma} \frac{\gamma!}{\gamma_1! \gamma_2! \cdots \gamma_n!} \cdot \frac{\partial^{\gamma} f(X_0)}{\partial x_1^{\gamma_1} \cdot \partial x_2^{\gamma_2} \cdots \partial x_n^{\gamma_n}} \cdot (dx_1)^{\gamma_1} (dx_2)$$

$$\cdot (dx_1)^{\gamma_1} (dx_2)^{\gamma_2} \cdots (dx_n)^{\gamma_n}$$

## Q.24

Show that a degenerate rectangle has zero content.

**Proof:-**

Let $R = [a_1, b_1] \times \cdots \times [a_n, b_n]$

and

$T = [a_1 - \epsilon, b_1 + \epsilon] \times \cdots \times [a_n - \epsilon, b_n + \epsilon]$

where $\epsilon > 0$.

Then $R \subset T$ M.M

since $a_i = b_i$ and $V(T) \leq 2\epsilon L^{n-1}$

where

$L = \max\{(b_j - a_j) \mid 1 \leq j \leq n\}$.

## Q.25

If $h(u,v) = f(u^2+v^2)$, then show that $vh_u - uh_v = 0$.

$h_u = 2uf'$ , $h_v = 2vf'$ M.M

Now

$vh_u = 2vuf'$ , $uh_v = 2uvf'$

So,

$$vh_u - uh_v = 2vuf' - 2uvf'$$
$$= 2f'(vu - uv)$$
$$= 2f'(0)$$
$$= 0$$